

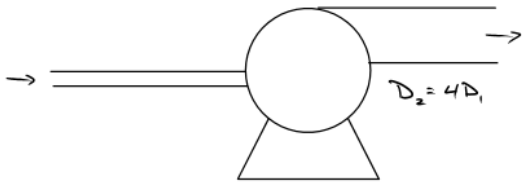
ME-ID Solution

"By initialing below I agree to the following: I will not discuss this exam with others who may be taking it at a different time due to schedule conflicts, and that I will not copy off of another student's exam, both of which are considered cheating. I understand that if I am caught cheating on this exam I will receive a score of zero."

Initials: Solution

Justify your answers mathematically or in words. Answers without justification will not receive credit.

1. (2 Points) Water flows through a narrow pipe into a pump. It exits through a pipe that has four times the diameter. Fill in the blank with a number to establish the relationship between the kinetic energy of the water leaving to the kinetic energy of the water entering: $KE_{\text{leaving}} = \underline{1/256} KE_{\text{entering}}$



$$V_1 \frac{\pi}{4} D_1^2 = V_2 \frac{\pi}{4} D_2^2$$

$$V_2 = \left(\frac{D_1}{D_2} \right)^2 V_1$$

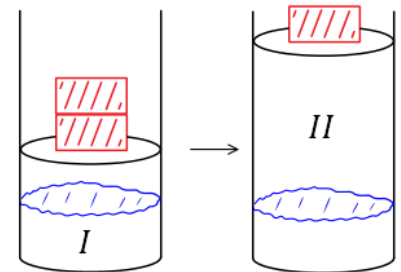
$$KE_1 = \frac{1}{2} \rho V_1^2$$

$$KE_2 = \frac{1}{2} \rho V_2^2$$

$$\frac{KE_2}{KE_1} = \left(\frac{V_2}{V_1} \right)^2 = \left(\frac{D_1}{D_2} \right)^4 = \left(\frac{1}{4} \right)^4 = 1/256$$

2. (2 Points) The piston-cylinder device moves between State I and State II by removing one brick. In both states the cylinder contains H_2O in vapor-liquid equilibrium. The pressure is halved, and the volume in State II becomes twice the volume in State I. Consequently, $P_I V_I = P_{II} V_{II}$.

- a. Heat was added to the system.
b. Heat was removed from the system.
c. The system remained adiabatic.
d. The state change is impossible.
e. The state change is possible, but more information is needed to answer.



The state change is possible, but it's hard to know whether heat would need to flow, and in what direction, unless you choose surrogate numbers from the water tables.

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S. Utrun

3. (2 Points) An electric motor drives a pump to move water through a pipe. The water exits as a free jet into the atmosphere. The pump continuously uses 10 kW of electrical energy, and its efficiency does not change. What would happen to the speed of the water at the inlet to the pump if the outlet pipe diameter is halved?

- a. It increases.
b. It remains the same.
c. It decreases.
d. More information is needed to answer.

If we assume that the pressure entering the pump, P_1 , is constant, then v_1 must decrease. However, P_1 might decrease due to the increased flow rate, and the exact value of P_1 would be needed in that case.

$$\eta \dot{W} = \Delta \dot{E}_{\text{mech}} = \dot{m} \left(\frac{1}{\rho} (P_2 - P_1) + \frac{1}{2} (v_2^2 - v_1^2) + g(z_2 - z_1) \right)$$

$$\eta \dot{W} = -\dot{m} \frac{P_1}{\rho} + \frac{1}{2} v_2^2 - \frac{1}{2} v_1^2$$

$$\dot{m} = \rho v_1 A_1$$

$$\eta \dot{W} = -\rho v_1 A_1 \frac{P_1}{\rho} + \frac{1}{2} v_2^2 - \frac{1}{2} v_1^2$$

$$\eta \dot{W} = \frac{1}{2} v_2^2 - \frac{1}{2} v_1^2 - v_1 A_1 P_1$$

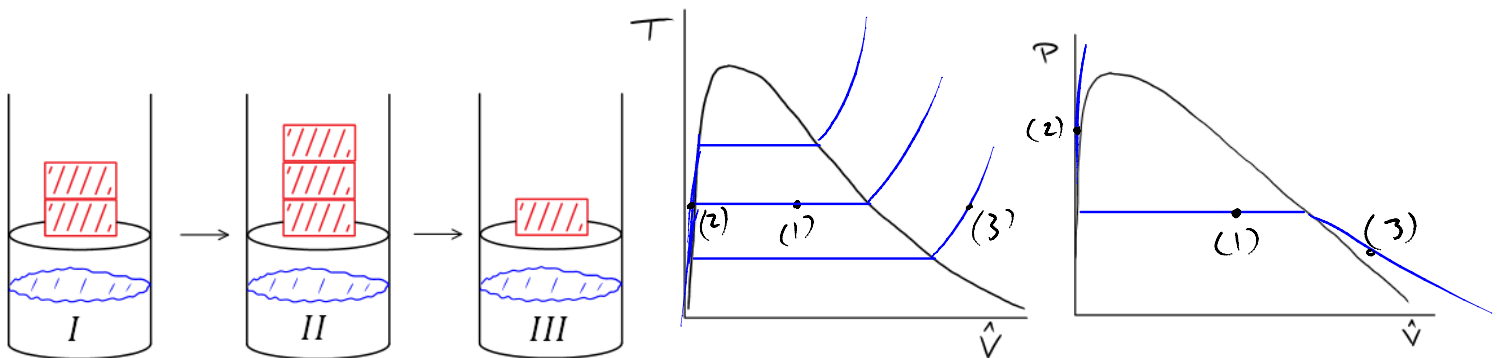
$$v_2 \uparrow \therefore v_1 \downarrow \text{ assuming } P_1 \text{ const.}$$

4. (4 Points) H₂O in vapor-liquid equilibrium is contained in a piston-cylinder device in State I. It is kept isothermal and goes through the following states. The heights of the pistons and the amount of liquid H₂O are not drawn to scale.

Plot the states on the P - $\hat{V}$ and T - $\hat{V}$ diagrams, and indicate the phase of the H₂O at State II and State III.

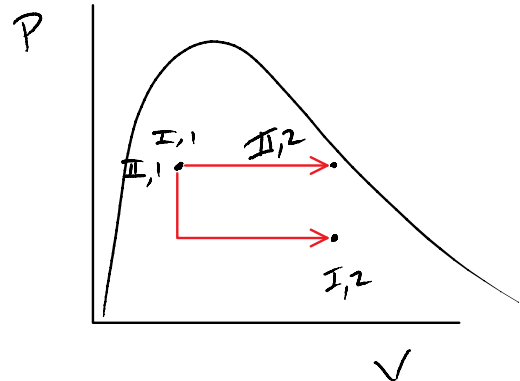
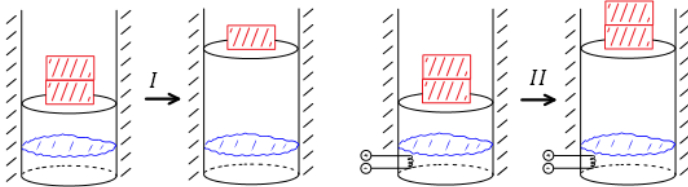
State II: [pressurized liquid / saturated liquid / vapor-liquid equilibrium / saturated vapor / superheated vapor]

State III: [pressurized liquid / saturated liquid / vapor-liquid equilibrium / saturated vapor / superheated vapor]



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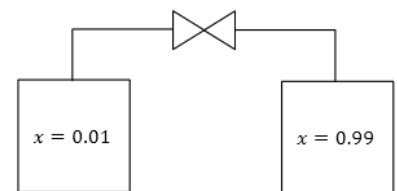
5. (2 Points) Two adiabatic piston-cylinder devices initially contain H_2O in vapor-liquid equilibrium. They have the same initial quality. One brick is removed from Device *I*. Device *II* is heated by an electric coil until it reaches the same final volume as Device *I*. How do the final qualities of the two devices compare?
- a. $x_I > x_{II}$ b. $x_I = x_{II}$ **c. $x_I < x_{II}$** d. More info is needed. e. It is not possible.



6. (2 Points) Heat is added isobarically to water with a quality of 0.25. Δu of the water is [less than / equal to / **greater than**] $c_v \Delta T$.

If the water remains under the two-phase envelope, then $\Delta T = 0$. However, $\Delta u > 0$, which is greater than $c_v \Delta T$. The answer would also be the same if the water was heated until it was superheated.

7. (2 Points) Two rigid vessels contain steam at 225°C . The vessels are connected by two pipes and a valve. Initially the valve is closed. Immediately after the valve is opened, the steam ____.
- a. rushes into the vessel on the left
b. rushes into the vessel on the right
c. does not move



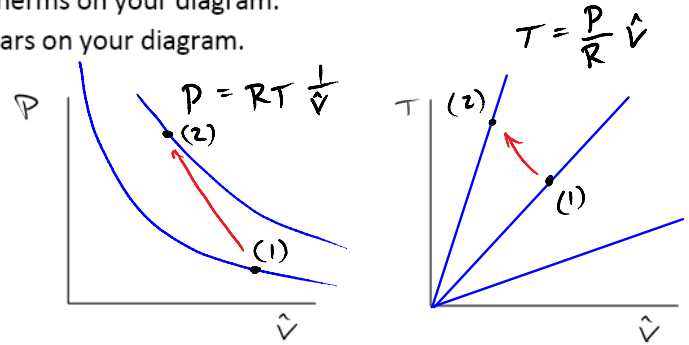
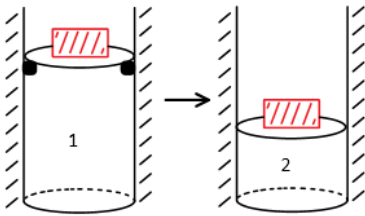
The pressure in both containers is the same. Therefore, no movement of steam occurs immediately after the valve is opened.

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8. (3 Points) The adiabatic piston-cylinder contains air. The stops are removed and the gas goes from State 1 to State 2.

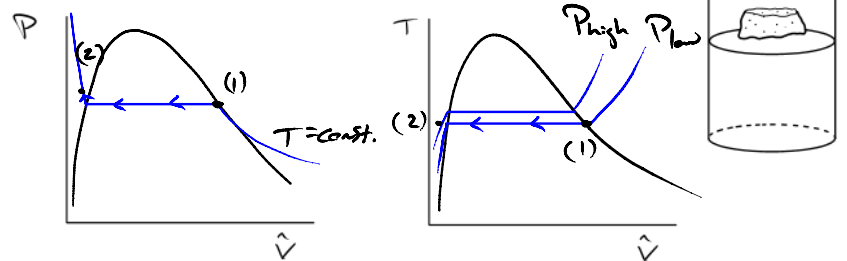
- Draw State 1 and State 2 on a P-V diagram. Include isotherms on your diagram.
- Draw State 1 and State 2 on a T-V diagram. Include isobars on your diagram.



The specific volume must decrease. The pressure must be halved, and the temperature must increase because work is done on the gas. Therefore, the gas must cross isotherms and isobars from State 1 to State 2.

9. (3 Points) The piston contains saturated water vapor. A pile of sand rests on top. A single grain of sand is then placed on top of the pile. The system remains isothermal. Draw the state transition on P-V and T-V diagrams.

- The water becomes a compressed liquid.
- The water becomes a saturated liquid ($x = 0$).
- The water becomes a mixture of liquid and vapor in equilibrium ($0 < x < 1$).
- The water becomes a superheated vapor.



10. (2 Points) The box is adiabatic and contains two halves: the left half contains air, and the right half contains a perfect vacuum. They are initially separated by a diaphragm, which then ruptures.

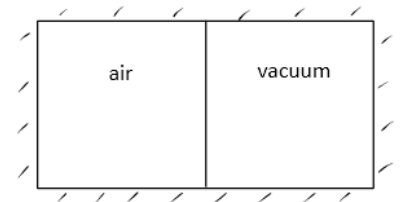
a. $P_2 < \frac{1}{2}P_1$

b. $P_2 = \frac{1}{2}P_1$

c. $P_2 > \frac{1}{2}P_1$

$$\frac{P_2 \hat{v}_2}{P_1 \hat{v}_1} = \frac{RT_2}{RT_1} = 1$$

$$P_2 = \frac{\hat{v}_1}{\hat{v}_2} P_1 = \frac{1}{2} P_1$$



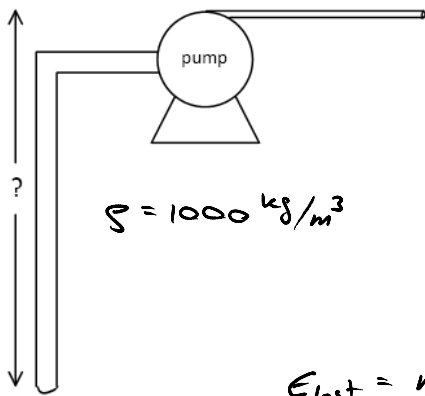
$$Q = \Delta u + W$$

$$\Delta u = C_v \Delta T$$

$$\rightarrow T_2 = T_1$$

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11. (20 Points) A pump moves water up from a well. The gage pressure of the water entering the pump is 50 kPa. The diameter of the pipe entering the pump is large. The water leaves the pump at a gage pressure of 150 kPa through a narrow pipe with a diameter of 10 cm. The motor driving the pump consumes 2.5 kW of electrical power and has an efficiency of 80%. The efficiency of the pump is 90%. Due to friction in the pipes, for every 1 kg of water flowing through the system 50 J of energy is lost. How deep can the well be if the pump is to move 10 kg/s of water?



$$\eta_{\text{pump}} \eta_{\text{motor}} \dot{W}_{\text{elec}} = \Delta \dot{E}_{\text{mech}} + \dot{E}_{\text{lost}}$$

6: energy entering the water is distributed between mechanical power and power lost due to friction

4: computing power lost due to friction

$$\dot{E}_{\text{lost}} = \dot{m} \cdot 50 \text{ J/kg} = 500 \text{ W} = 0.5 \text{ kW}$$

5: simplifying mechanical energy equation

$$\Delta \dot{E}_{\text{mech}} = \dot{m} \left(\frac{1}{\rho} (P_2 - P_1) + \frac{1}{2} (v_2^2 - v_1^2) + g(z_2 - z_1) \right)$$

$$v_2 \frac{\pi}{4} D_2^2 = \frac{\dot{m}}{\rho} \longrightarrow v_2 = \frac{\dot{m}}{\rho} \frac{4}{\pi D^2} = 1.273 \text{ m/s}$$

4: solving for v_2

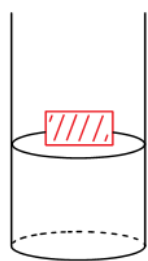
$$z_2 = \frac{1}{g} \left(\frac{1}{\dot{m}} (\eta_{\text{pump}} \eta_{\text{motor}} \dot{W}_{\text{elec}} - \dot{E}_{\text{lost}}) - \frac{1}{\rho} (P_2 - P_1) - \frac{1}{2} v_2^2 \right) = 3.0 \text{ m}$$

1: correct answer

12. (30 Points) The 1428 kg brick sits on top of the piston, which has an area of 0.01 m^2 . The piston is weightless.

Atmospheric pressure of 100 kPa surrounds the device. In State 1, the cylinder contains 0.5 m^3 of saturated liquid H_2O . The brick is then removed, and the H_2O expands isothermally until it reaches State 2. The piston then moves downward until it reaches State 3, at which point the H_2O is at its original volume. Make sure you are working in absolute pressure and that your units are okay. Don't interpolate, just use the nearest values.

- Draw the three states on the diagrams. Connect them appropriately.
- Between States 1 and 2, was heat added to or removed from the H_2O , or was the system adiabatic during this step? If it was not adiabatic, indicate whether heat was added or removed, and calculate the amount, in kJ.
- Answer the same question for the transition between States 2 and 3.
- Work was done on the H_2O by the atmosphere between States 2 and 3. Shade in the area on the graph that corresponds to the amount of work that it did.



$$P_1 = \frac{M_{\text{brick}} g}{A} + P_{\text{atm}} = 1500 \text{ kPa}$$

$$= \frac{1428 \text{ kg} \cdot 9.81 \text{ m/s}^2}{0.01 \text{ m}^2} \cdot 0.001 \frac{\text{kPa}}{\text{Pa}} + 100 \text{ kPa}$$

3: necessary properties for State 1

$$\left. \begin{array}{l} P_1 \approx 1500 \text{ kPa} \\ x_1 = 0 \end{array} \right\} \begin{array}{l} u_1 \approx 842.82 \text{ kJ/kg} \\ \hat{v}_1 \approx 0.001154 \text{ m}^3/\text{kg} \\ T_1 \approx 200^\circ\text{C} \end{array}$$

$$\left. \begin{array}{l} P_2 = 100 \text{ kPa} \\ T_2 \approx T_1 \end{array} \right\} \begin{array}{l} u_2 \approx 2658.2 \text{ kJ/kg} \\ \hat{v}_2 \approx 2.1724 \text{ m}^3/\text{kg} \end{array}$$

3: necessary properties for State 2

$$P_3 = 100 \text{ kPa}$$

$$\hat{v}_3 = \hat{v}_1 = 0.001154 \text{ m}^3/\text{kg}$$

$$x_3 = \frac{\hat{v} - \hat{v}_f}{\hat{v}_g - \hat{v}_f} = \frac{0.001154 - 0.001043}{1.6941 - 0.001043} = 6.6 \times 10^{-5}$$

$$u_3 = u_f + x_3 u_{fg} = 417.40 + 6.6 \times 10^{-5} \cdot 2088.2 = 417.5 \text{ kJ/kg}$$

4: calculating u_3 using properties

$$m = \frac{V_1}{\hat{v}_1} = \frac{0.5 \text{ m}^3}{0.001154 \text{ m}^3/\text{kg}} = 433 \text{ kg}$$

3: calculating m

$$Q_{12} = m(u_2 - u_1) + m P_2 (\hat{v}_2 - \hat{v}_1) = 433 (2658.2 - 842.82) + 433 \cdot 100 (2.1724 - 0.001154)$$

$$= 880000 \text{ kJ added}$$

5: energy balance for State 1 to 2

1: both answers correct

$$Q_{23} = m(u_3 - u_2) + m P_2 (\hat{v}_3 - \hat{v}_2) = 433 (417.5 - 2658.2) + 433 \cdot 100 (0.001154 - 2.1724)$$

$$= -1100000 \text{ kJ removed}$$

5: energy balance for State 2 to 3

